Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

JAVA ASSIGNMENT

NAME:Kavya Gada

SAP ID:60017230102

ROLL NO.:A018

BRANCH:AIML

BATCH:A1

DATE OF SUBMISSION:28/03/24

NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018

Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

JAVA :- Experiment No:3

- AIM :- To implement strings

* Description of experiments

i) explain strings in Java

→ In Java, string is basically an object that represents sequence of ~~e that~~ char values. An array of characters works same as Java string. Java string class provides a lot of methods to perform operation on strings such as compare(), concatenate(), equals(), length(), replace(), compareTo(), substring() etc. Java strings are created and ~~many~~ manipulated through the string class. Once created, string is immutable i.e. its value cannot be changed.

ii) Difference in declaring the string with and without new keyword (string literal).

String object	String literal
1. Set of character that is created using the new() operator	1. Set of characters that is created by enclosing them inside a pair of double quotes
2. Object is created in the memory whereas its reference will be pointed to string pool.	2. Here, the reference will be directly referred to string pool. JVM initially checks for the availability with same value is string constant pool.

FOR EDUCATIONAL USE

NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018

Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

For eg:-
String s = "Hello world"

For eg:-
String s = new String ("Hello world")

iii) String Constant pool and Heap.

→ String Constant pool
String pool is nothing but a storage area in Java heap where String literals stores. It is also known as string Intern pool or string Constant pool. It is just like object allocation. By default, it is empty and privately maintained by the Java String class. String objects created without the use of new keyword are stored in the string Constant pool is that it does not create the same string objects if there is already string constant in the pool.

Heap Memory :-

Heap memory is a run time data area from which the memory for all java class instances and arrays is allocated. The heap is created when the JVM starts-up and may increases or decrease in size while the application runs. The heap can be of fixed size or variable size depending on the garbage collection strategy.

Sundaram
FOR EDUCATIONAL USE

NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018

iv) Different string classes and their methods for handling strings in java

→ The java.lang.String class provides a lot of built-in methods that are used to manipulate string in java. By the help of these methods, we can perform operations on string objects such as trimming, concatenating, converting, comparing, replacing string etc.

- toUpperCase() and toLowerCase() method
  The java String toUpperCase() method converts the string into Upper case letter and String toLowerCase() method into Lower case letter.

- trim() method
  The String class trim() method eliminates white spaces before and after the string.

- Starts with() and ends with() method
  Starts with() check whether the string start with the letter passed as arguments and ends with() method check whether the string ends with the letter passed as argument.

- CharAt() Method
  The String class CharAt() method return a character at specific index

- Length() Method :- The String class length() method returns length of the specific string

FOR EDUCATIONAL USE

[Type here]

- Value of () Method
  The String class value of () method and converts given type such as int, long, float, double, boolean, char and char array into String.

- replace () Method
  The String class replace () method replace all occurence of first sequence of characters with second sequence of characters.

* Difference between String, String Buffer, String Builder and String Tokenizer classes

	String	String Buffer	String Builder	String Tokenizer
Storage	String pool	Heap	Heap	Heap
Mutability	Immutable	Mutable	Mutable	–
Performance	Slow	faster than String	faster than String Buffer	faster than String Builder
Thread Safe	Not used in threaded environment	Multi-thread environment	Single-Thread environment	–
Definition	objects used to represent sequence of characters	peer class of string that creates mutable string	string objects that can be modified	It can break a string into tokens

Sundaram
FOR EDUCATIONAL USE

NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018

Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

* Conclusion :- I learnt how to implement string using string class and its various methods

3. c) Write a program to check if a string is a Pallindrom or not using in-built fxn.

Program:-

```
class Strings {
public static void main (String args[]) {
 int result;
String Buffer sb = new String Buffer ("nitin");
String Buffer SC = new String Buffer ();
 Sc = sb;
String SC = sc.toString ();
Sb = sb.reverse ();
String sd = sb.toString();
result = sd.compareTo (se);
System.out.println (result);
if (result == 0)
{
 System.out.println ("pallindrome");
}
 else {
 System.out.println (" not pallindrome");
 }
}
}
```

Sundaram
FOR EDUCATIONAL USE

SAP ID: 60017230102

ROLL NO.:A018

[Type here]

#output:-

0

Pallindrome.

14/3/24

Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

Experiment No: 3

To implement Strings

WAP to find out number of uppercase & lowercase characters, blank spaces and digits from the string.

```
import java.util.Scanner;

public class StringParsing
{
	public static void main(String[] args)
	{
		//Scanner is a class used to get the output from the user
		Scanner sc=new Scanner(System.in);
		//Input the string
		System.out.println("Enter the string!");
		String s=sc.nextLine();
		//counter to count the uppercase and lowercase letters
		int uppercase=0,lowercase=0, digit=0,blankSpace=0, specialCharacter=0;
		//looping until the string length is zero
		for(int i=0;i<s.length();i++)
		{
			if(Character.isLowerCase(s.charAt(i)))
			{
				lowercase++;
			}
			else if(Character.isUpperCase(s.charAt(i)))
			{
				uppercase++;
			}
			else if(Character.isDigit(s.charAt(i)))
			{
				digit++;
			}
```


NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018

```
                else if(s.charAt(i)==' ')
                {
                        blankSpace++;
                }
                else
                {
                        specialCharacter++;
                }
        }
        //Print the output
        System.out.println("No. of lowercase letter : " + lowercase);
        System.out.println("No. of uppercase letter : " + uppercase);
        System.out.println("No. of Digits : " + digit);
        System.out.println("No. of Blank Spaces : " + blankSpace);
        System.out.println("No. of Special Characters: " + specialCharacter);
        //to give a space of one line
        System.out.println();
    }

}
```

```
C:\kavya\java\College Pratice Q\Exp 3 Code>java StringParsing
Enter the string!
abcd_1987@gmail.com
No. of lowercase letter : 12
No. of uppercase letter : 0
No. of Digits : 4
No. of Blank Spaces : 0
No. of Special Characters: 3


C:\kavya\java\College Pratice Q\Exp 3 Code>
```


NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018

WAP to count the frequency of occurrence of a given character in a given line of text.

```
import java.util.Scanner;

public class CharacterFrequency
{
	public static void main(String[] args)
	{
		//Scanner is a class used to get the output from the user
		Scanner sc=new Scanner(System.in);
		//Input the string
		System.out.println("Enter the string!");
		String s=sc.nextLine();
		//counter to count the uppercase and lowercase letters
		int frequency=0;
		System.out.println("Enter the Character to find its frequency: ");
		char ch=sc.next().charAt(0);
		//looping until the string length is zero
		for(int i=0;i<s.length();i++)
		{
			if(s.charAt(i)==ch)
				frequency++;
		}
		//Print the output
		System.out.println("The Character"+ ch+ "has occurred "+frequency+" Times");
	}

}
```


NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018

```
C:\kavya\java\College Pratice Q\Exp 3 Code>javac CharacterFrequency.java

C:\kavya\java\College Pratice Q\Exp 3 Code>java CharacterFrequency
Enter the string!
ENGINEERING
Enter the Character to find its frequency:
E
The CharacterEhas occurred 3 Times

C:\kavya\java\College Pratice Q\Exp 3 Code>
```

WAP to check if a string is a palindrome or not using inbuild functions.

```
import java.util.Scanner;

public class StringPalindrome
{
	public static void main(String[] args)
	{
		//Scanner is a class used to get the output from the user
		Scanner sc=new Scanner(System.in);
		//Input the string
		System.out.println("Enter the string!");
		String s=sc.nextLine();
		StringBuffer sb=new StringBuffer(s);
		sb=sb.reverse();
		String rev=sb.toString();//converting stringBuffer object ot string object
		if(s.compareTo(rev)==0)
		{
			System.out.println("String+ "+s+" is Palindrome");

		}
		else
		{
```


NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018

```
                System.out.println("String+ "+s+" is Not Palindrome");

            }

        }

}
```

```
C:\kavya\java\College Pratice Q\Exp 3 Code>javac StringPalindrome.java

C:\kavya\java\College Pratice Q\Exp 3 Code>java StringPalindrome
Enter the string!
RacecaR
String+ RacecaR is Palindrome

C:\kavya\java\College Pratice Q\Exp 3 Code>
```


NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018